Data Science

Statistics | Hypothesis Testing

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Data Science Track

Hypothesis Testing

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LEARNING OBJECTIVE

In this chapter, we'll explore hypothesis testing for population proportions, differences in proportions, population means, and differences in means.

By the time you reach the end of this chapter, you will be able to:

1. Describe the historical development of hypothesis testing, including Ronald Fisher's contributions.
2. Differentiate between mean (μ) and standard deviation (σ or s) as measures of central tendency and dispersion.
3. Interpret the concept of the p-value and its role in hypothesis testing.
4. Explain the Hypothesis Testing Process.
5. Apply the Steps of Hypothesis Testing
6. Understand the Concept of Hypothesis Testing for Population Proportions
7. Define and Apply Key Statistical Formulas
8. Formulate and Interpret Hypotheses
9. Recognize and Avoid Common Mistakes
10. Understand the Purpose of Comparing Population Proportions
11. Apply Key Statistical Concepts and Formulas

GETTING STARTED

Our researcher friend, Lina, used confidence intervals to make inferences about the population of university students' heights. Today, she's more interested in their dining habits.

"University students: 80% attending for the food, 20% pretending to attend lectures."

No worries, it's not biased data—just a bit of fun!

From her sample of 150 students, she observed some patterns:

- Many students spend over 10 JOD per meal, prompting Lina to ask, "Do university students spend significantly more than 10 JOD on average?"
- She notices differences in spending between male and female students and wonders, "Is the difference in average spending statistically significant?"

12. Interpret Hypothesis Testing Results
13. Recognize Real-World Applications of Population Proportion Testing
14. Explore Extensions Beyond Traditional Hypothesis Testing
15. Understand the Concept of Hypothesis Testing for a Population Mean
16. Apply Key Statistical Formulas
17. Formulate and Interpret Hypotheses
18. Understand the Purpose of Comparing Means Between Two Groups
19. Apply Key Statistical Concepts
20. Calculate the Test Statistic and p-value
21. Interpret Hypothesis Testing Results
22. Understand the Role of Regression in Hypothesis Testing
23. Identify dependent (outcome) and independent (predictor) variables in a regression model and their roles.
24. Understand Multicollinearity and Its Impact
25. Apply Regression Analysis to Control for Confounding Variables
26. Interpret Regression Coefficients and p-values
27. Explore the Role of Confidence Intervals in Feature Selection

- Finally, Lina is curious if male students are more likely to eat out regularly compared to female students.

To answer these questions, Lina turns to hypothesis testing, a framework for evaluating claims about a population based on sample data.

Hypothesis Testing

Behind the Formula


" At the end, it's not the same."

The story of hypothesis testing is a tale of intellectual battles in statistics. It was developed by **Ronald Fisher**, who introduced the concept of the p-value in the 1920s. Fisher's work on agricultural experiments led to the method of testing whether observed effects were due to chance.

The tea-tasting experiment was conducted by him. Fisher is famously detailed in his book *The Design of Experiments* (1935). He designed this test to assess whether a lady could distinguish between tea prepared in two different ways: milk poured before tea vs. tea poured before milk.

Key Concepts

- **Mean (μ) and Standard Deviation (σ or s)** – Measures of central tendency and dispersion, respectively.
- **p-value** – The probability of obtaining a test statistic as extreme as the observed one, assuming the null hypothesis (H_0) is true.

1.1Hypothesis Testing

Hypothesis testing is a structured method for evaluating claims about a population parameter based on sample data. It provides a systematic way to decide whether the observed evidence supports a given assumption or contradicts it.

The process involves two opposing statements

Null Hypothesis (H_0)	Alternative Hypothesis (H_a)
Represents the status quo or no effect. It is the claim we test against.	Represents what we want to prove or verify.

Example:

If we are testing whether the average daily meal cost of university students exceeds 10 JOD, the hypotheses could be:

H_0: μ = 10 (The average meal cost is 10 JOD.)

H_a: μ > 10 (The average meal cost is greater than 10 JOD.)

"Budget for books: 5 JOD, budget for food: 10 JOD" ... Priorities? Fully optimized!

Note:

The alternative hypothesis can be **one-tailed** or **two-tailed**. The two-tailed alternative hypothesis for the above example would be ($\mu \neq 10$). However, each of ($\mu > 10$) and ($\mu < 10$) are examples of one-tailed alternatives.

One-Tailed vs. Two-Tailed Tests

One-tailed tests check for an increase or decrease, while two-tailed tests check for any difference.

1.2 How Hypothesis Testing Works

1. **Set a Significance Level (α)**: This represents the probability of rejecting H_0 when it is actually true (a Type I error). Common values for α are 0.05 or 0.01.
2. **Calculate a Test Statistic**: The test statistic measures how far the sample data deviates from the null hypothesis. It is calculated as:

$$Test\ Statistic = \frac{Observed\ Value - Hypothesized\ Value}{Standard\ Error}$$

3. **Determine the p-value**: The p-value represents the probability of observing a test statistic as extreme as the one calculated, assuming H_0 is true. The p-value is calculated from the test statistic.
4. **Decision Rule**:
   - If p-value < α, reject H_0.
   - If p-value ≥ α, fail to reject H_0.

In the examples of this chapter, we will not calculate p-values or compare them to α. Instead, we will only show the Null and Alternative hypotheses for each example. *Actual hypothesis testing will be performed using Python in the exercises.*

Math in AI

Statistical tests evaluate whether an AI model exhibits gender, racial, or other biases by comparing group-level predictions.

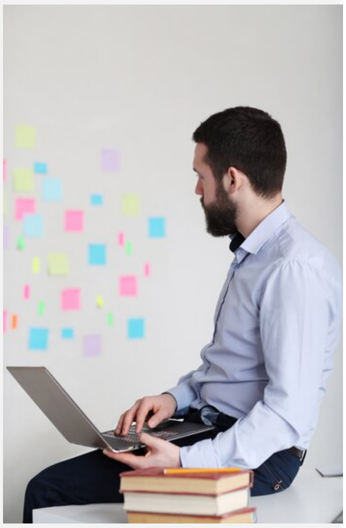
Imagine an AI model used to screen job applicants. A hypothesis test is conducted to check if **male and female** candidates receive interview invitations at the same rate.

H_0: The AI selects male and female candidates at the same rate.

H_a: There is a significant difference in selection rates between genders.

Verify Your Learning!

1. Explain the role of hypothesis testing in statistical analysis. How does it help in making data-driven decisions? Provide an example from a real-world scenario where hypothesis testing can be used to assess a claim about a population parameter.
2. A company wants to test whether the average time employees spend on professional development exceeds 5 hours per week. Formulate the null and alternative hypotheses for this scenario. Describe the steps required to conduct the hypothesis test, including setting a significance level, calculating a test statistic, and making a decision based on the results.



2. Hypothesis Testing for Population Proportion

When testing claims about a single population proportion, hypothesis testing determines whether the observed proportion differs significantly from a hypothesized value; either in increase, decrease, or both (two-tailed).

The Standard Error in this case is:

$$SE = \sqrt{\frac{p_0(1-p_0)}{n}}$$

And the test statistic is:

$$test\ statistic = \frac{\hat{p}-p_0}{SE}$$

Where:

- $\hat{p}$: Observed sample proportion
- p_0: Hypothesized proportion
- n: Sample size

Example

Lina believes that 50% of students eat out at least once a day. However, from her sample, she finds this proportion to be 60%. Lina wants to know if the observed proportion is statistically significantly greater than what she had believed (Considering a significance level of 0.05).

- H_0: p = 0.5
- H_a: p > 0.5

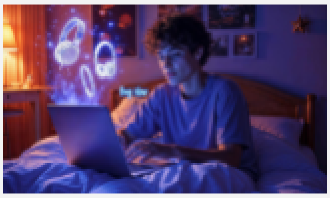
After calculating the test statistic and p-value, if the p-value is less than the significance level (0.05) we reject the null hypothesis in favor of the alternative. Otherwise, we fail to reject the null hypothesis.

mistake spotting

Learners often get confused with two-tailed tests and incorrectly interpret the results. In two-tailed tests, the p-value should be interpreted as the probability of obtaining a result as extreme as the observed one, in either direction.

Verify Your Learning!

1. Explain the role of the null and alternative hypotheses in hypothesis testing for a population proportion. Discuss how the significance level and p-value are used to determine whether to reject or fail to reject the null hypothesis.
2. A survey claims that 40% of customers prefer shopping online rather than in physical stores. To validate this claim, a researcher collects a random sample of 200 customers and finds that 90 of them prefer online shopping. Conduct a hypothesis test at a 5% significance level to determine if there is sufficient evidence to reject the claim. Clearly state the hypotheses.



3. Hypothesis Testing for Difference in Population Proportions

This type of test is usually used to answer the question: **is there a statistically significant difference in proportion between two groups.**

This is usually achieved by setting the null hypothesis to be that the difference in proportions is equal to zero (which means that there is no significant difference in proportions), and the alternative hypothesis that the difference in proportions is not equal to zero (such that one proportion is greater than the other).

The standard error here is:

$$SE = \sqrt{\hat{p}(1 - \hat{p})(\frac{1}{n_1} + \frac{1}{n_2})}$$

Where $\hat{p}$ is the pooled sample proportion:

$$\hat{p} = \frac{x_1+x_2}{n_1+n_2}$$

And the test statistic is calculated as the observed difference in proportions minus the hypothesized difference (zero), divided by the standard error:

$$test\ statistic = \frac{\left(\hat{p}_1-\hat{p}_2\right)-(0)}{SE}$$

Example

Lina finds that out of all males in her sample, 65% of them eat out regularly. For females, the proportion is 60%. She asks, is there a statistically significant difference between the two proportions between the two groups (considering a significance level of 0.05)?

- H_0: $p_m - p_f = 0$
- H_a: $p_m - p_f \neq 0$

After calculating the test statistic and p-value, we again compare the p-value to the significance level (0.05). We reject the null hypothesis if p-value is less than 0.05; i.e. there is a statistically significant difference. Otherwise, we fail to reject the null hypothesis; so there is no statistically significant difference.

Brain Power

Imagine you're tasked with analyzing the differences in proportions of two groups, but you also have access to a rich set of other variables (e.g., age, income, location). How might you go beyond traditional hypothesis testing and incorporate these additional factors into your analysis? What alternative methods or models could you use to uncover deeper insights, and how might this shift your perspective on what constitutes a "significant" difference? Consider the potential of combining hypothesis testing with machine learning or other advanced techniques to capture complexity in your data.

Verify Your Learning!

1. Why is the pooled sample proportion used in hypothesis testing for the difference in population proportions? Explain how the null and alternative hypotheses are formulated in this test, and describe a real-world scenario where this type of test would be applicable.
2. A marketing team wants to compare the effectiveness of two different ad campaigns (A and B) in attracting customers. From a sample of 300 customers who viewed campaign A, 150 made a purchase. From a sample of 250 customers who viewed campaign B, 110 made a purchase. Conduct a hypothesis test at a 5% significance level to determine if there is a statistically significant difference in the conversion rates of the two campaigns. Clearly state the hypotheses.

4. Hypothesis Testing for Population Mean

This test evaluates whether the sample mean differs from a hypothesized population mean.

It is used when we have sample data to infer about the population, the population standard deviation is unknown, and the sample size is large or the sample is from a normal distribution.

The standard error for this calculation is:

$$SE = \frac{s}{\sqrt{n}}$$

Where :

- **s**: observed standard deviation
- **n**: sample size

And the test statistic is the observed sample mean minus the hypothesized population mean, divided by the standard error:

$$test\ statistic = \frac{\bar{x} - \mu_0}{SE}$$

Hypothesis Testing for Population Mean is like being a judge, where you use sample data to decide if the population mean is guilty of not measuring up!

Example

Lina wants to test whether students spend on average more than 10 JOD per meal.

- H_0: $\bar{x} = 10$
- H_a: $\bar{x} > 10$

As before, we calculate the test statistic and the p-value. Then we compare the p-value to the significance level and make a decision on whether we will reject the null hypothesis.

Verify Your Learning!

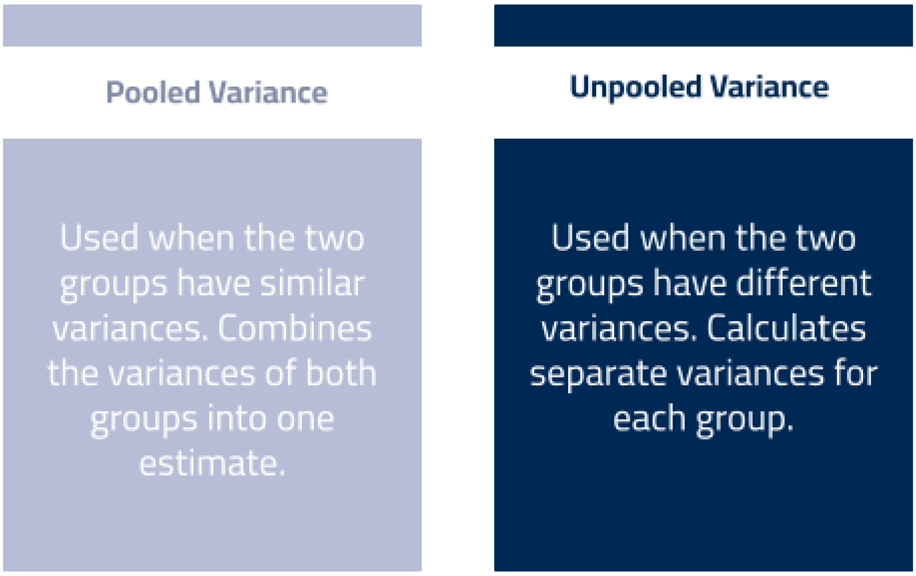
1. Explain the importance of hypothesis testing for a population mean. In your response, discuss why the standard error is used in the test statistic formula and how sample size influences the reliability of the test results. Provide an example of a situation where a two-tailed test would be more appropriate than a one-tailed test.
2. A nutritionist believes that the average daily sodium intake of adults in a city is different from the recommended 2,300 mg. To test this, she collects a random sample of 50 adults and finds that their mean sodium intake is 2,450 mg, with a standard deviation of 400 mg. Conduct a hypothesis test at a 5% significance level to determine whether the sample provides enough evidence to conclude that the mean sodium intake differs from the recommended amount. Clearly state the hypotheses.

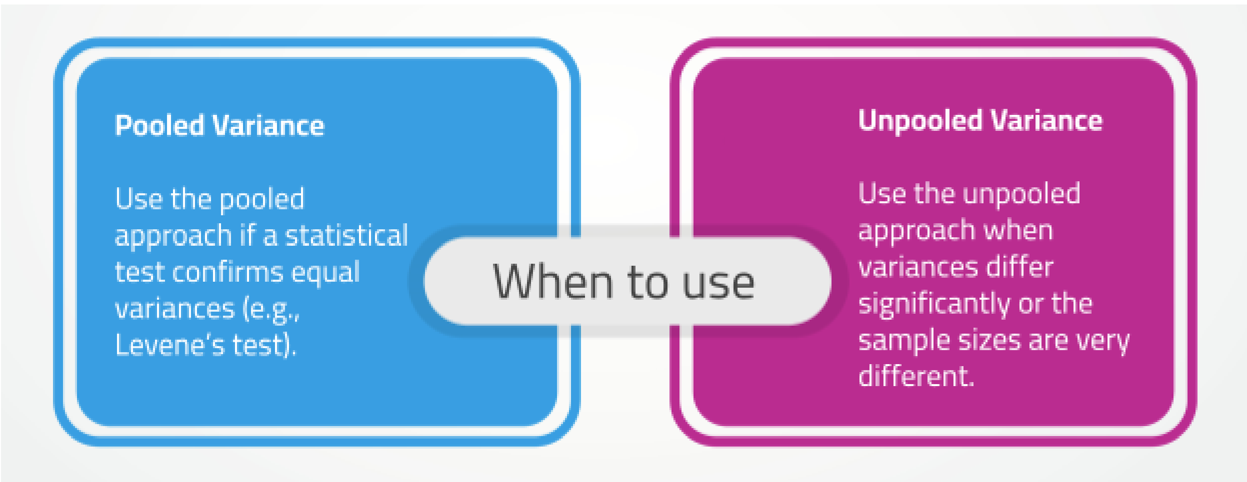

"It will always need a little bit of salt"

5. Hypothesis Testing for Difference in Means

This test compares whether two groups differ significantly in their means.

Pooled vs. Unpooled Variance





Example

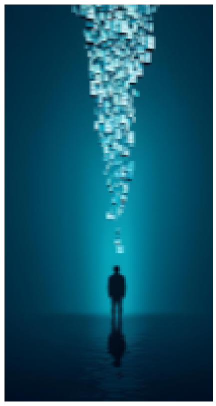
Lina wants to test if there is a statistically significant difference in average spending per meal between males and females (considering 0.05 significance level).

- H_0: $\bar{x}_1 - \bar{x}_2 = 0$ (no difference in means)

- H_a: $\overline{x}_1 - \overline{x}_2 \neq 0$ (means are different)

Then we decide on the pooled vs unpooled approach. Based on the chosen method, we calculate the test static and the p-value. By comparing the p-value to the significance level, we make a decision on the null hypothesis.

Brain power



Statistical tests help us determine whether differences between groups are significant—but what if the definition of "difference" itself is more complex than just a mean comparison? Could two groups have the same average but entirely different distributions? How might looking beyond the mean—examining variability, skewness, or even underlying data patterns—change the story hidden within the numbers?

Verify Your Learning!

1. Explain the difference between pooled and unpooled variance in hypothesis testing for the difference in means. How does the decision to use pooled or unpooled variance impact the test results? Provide a real-world example where an unpooled variance approach would be more appropriate than a pooled variance approach.
2. A researcher wants to compare the average test scores of two groups of students: one taught using traditional methods and the other using an AI-based learning platform. The first group (n_1 = 30) has a mean score of 78 with a standard deviation of 10, while the second group (n_2 = 35) has a mean score of 85 with a standard deviation of 12.

   Conduct a hypothesis test at a 5% significance level to determine whether there is a significant difference in mean test scores. Clearly state the hypotheses, determine whether pooled or unpooled variance should be used.

6. Important Notes on Hypothesis Testing

For hypothesis testing to produce valid and reliable results, certain assumptions must be met. These assumptions ensure that the test statistic follows the expected distribution, making the conclusions statistically sound.

6.1 Assumptions for Proportion Tests (One and Two Population Proportions)

✔ Random Sampling: The sample should be selected randomly to avoid bias.

✔ Sample Size Condition: The sample size should be large enough for the normal approximation to apply. A general rule is:

$$np_0 \geq 10, \;\; and \;\; n(1 - p_0) \geq 10$$

for one-proportion tests, and

$$n_1 p_1, \;\; n_1(1 - p_1), \;\; n_2 p_2, \;\; n_2(1 - p_2) \geq 10$$

or two-proportion tests. This ensures that the sampling distribution of the sample proportion is approximately normal.

✔ Independence of Observations: Observations should not influence each other. If sampling is done without replacement, the population should be at least 10 times larger than the sample size ($N \geq 10n$) to maintain independence.

✔ Independent Groups (for Two Proportions Tests): The two groups being compared should be independent of each other.

6.2 Assumptions for Mean Tests (One and Two Population Means)

Random Sampling is like picking names out of a hat – everyone has an equal shot, so no one can say you played favorites!

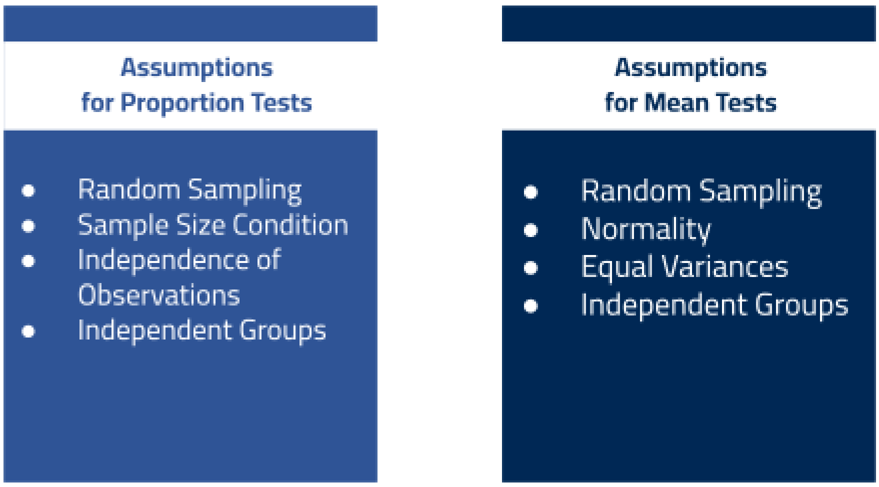
✔ Random Sampling: The sample must be randomly selected to ensure unbiased estimates.

✔ Normality:

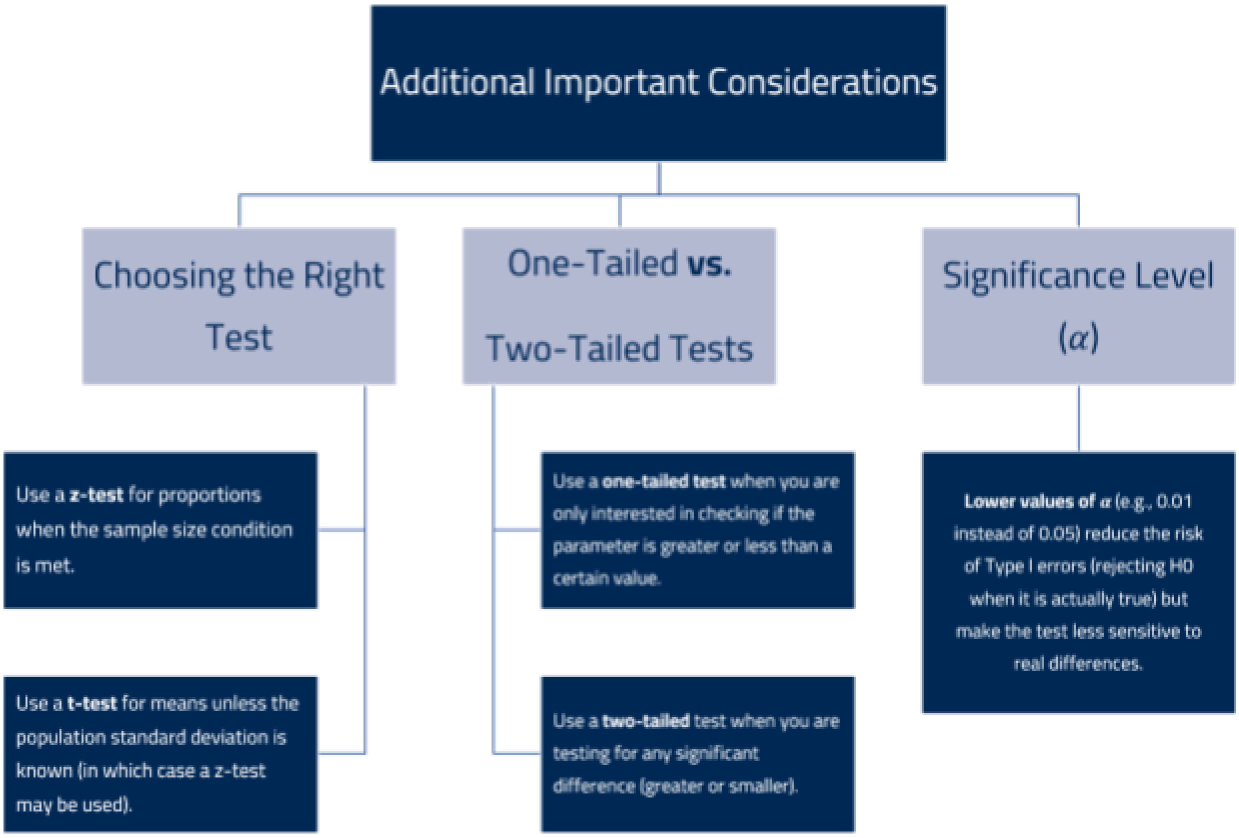
- If the sample size is small ($n < 30$), the population distribution should be approximately normal.
- If the sample size is large ($n \geq 30$), the Central Limit Theorem ensures that the sample mean follows a normal distribution, even if the population distribution is not normal.

✔ Equal Variances (for Pooled Two-Sample t-Test): If using the pooled variance approach, the variances of the two groups should be roughly equal. This can be tested using Levene's test or by comparing standard deviations. If variances are unequal, the unpooled (Welch's t-test) method should be used.

✔ Independent Groups (for Two Means Tests): The two groups should be independent of each other, meaning that an observation in one group should not influence an observation in the other group.



6.3 Additional Important Considerations



6.4 Parametric vs. Non-Parametric Tests

Hypothesis tests are generally classified as parametric or non-parametric, depending on the assumptions they make about the data distribution.

- **Parametric Tests**: Parametric tests assume that the population data follows a specific distribution (usually normal) and rely on parameters like the mean and standard deviation. These tests are more powerful when their assumptions are met.

  Examples:

  - z-test (for large samples and known population variance)
  - t-test (for small samples or unknown population variance)
  - ANOVA (for comparing means of multiple groups)

- **Non-Parametric Tests**: Non-parametric tests do not assume a specific population distribution. They are useful when the normality assumption is violated, the data contains outliers, or the measurement scale is ordinal. While they are more flexible, they tend to be less powerful than parametric tests when normality holds.

Therefore, if the normality assumption is met, parametric tests are preferred because they are more powerful. If normality is questionable or the data is ordinal/categorical, non-parametric tests are a safer choice.

Non-Parametric Tests: The chill, go-with-the-flow tests that don't care about fancy assumptions but might take a bit longer to get the job done.

Hypothesis Testing in Fraud Detection

Financial institutions use hypothesis testing to detect fraud in credit card transactions. By modeling legitimate transactions, they set thresholds for unusual spending behavior. If a transaction significantly deviates from the expected pattern, an alert is triggered.

For example:

H_0: The transaction is normal

H_a: The transaction is fraudulent If the test statistic falls in the rejection region, the system flags the transaction for review.

Verify Your Learning!

1. Explain the key assumptions required for hypothesis testing of population proportions and means. Why are these assumptions important, and how do they impact the validity of test results? Additionally, differentiate between parametric and non-parametric tests, providing an example of when a non-parametric test would be more appropriate than a parametric test.
2. A researcher is conducting a hypothesis test to compare the mean salaries of employees in two companies. She collects a random sample of 20 employees from each company and finds that the salary distributions are highly skewed. Additionally, Levene's test indicates that the variances between the two groups are significantly different. Based on these findings, what statistical test should she use, and why? Discuss the reasoning behind your choice, considering assumptions of parametric and non-parametric tests.

How This Concept Shapes My Field

Regression Analysis for Hypothesis Testing

Math in Situation


" you will always say YES to lunch with friends"

Lina has been using hypothesis testing to analyze university students' dining habits. She compared the average meal expenses of male and female students and found a statistically significant difference. However, something was bothering her—could gender alone explain the variation in spending, or were other factors at play?
She decided to dig deeper. Perhaps meal cost is also influenced by income level, dining frequency, or group size. If she only focuses on gender, she risks oversimplifying the analysis. What if males tend to eat out more frequently, and that's the real reason they spend more? To answer this, Lina turns to regression analysis, which allows her to compare means while controlling for other variables.

Key Concepts

- **Dependent vs. Independent Variables:** The dependent variable (y) is what we aim to predict, while independent variables (x) are the factors influencing it.
- **Multicollinearity:** A situation where independent variables are highly correlated, which can distort regression estimates.

From Univariate to Multivariate Analysis

Hypothesis testing for means is a univariate analysis, meaning it examines one factor at a time. When comparing two groups (e.g., male vs. female spending), it does not account for other influencing factors. This limitation can lead to misleading conclusions if an observed difference is actually caused by another variable.

Regression analysis solves this problem by treating the outcome (meal cost) as the dependent variable (target) and including multiple independent variables (predictors) to explain it. In this way, we can isolate the effect of each variable while holding the others constant.

Regression analysis
The simplest form of regression, involving a straight-line relationship between the independent and dependent variables.

Regression for Hypothesis Testing

A linear regression model can be written as:

$$y = w_0 + w_1 x_1 + w_2 x_2 + \dots + w_n x_n$$

Where:

- y : dependent variable (e.g., meal cost)
- $x_1, x_2, \dots, x_n$: independent variables (e.g., gender, income, dining frequency)
- w_0 : intercept
- $w_1, w_2, \dots, w_n$: coefficients measuring the effect of each variable

Each predictor in the regression has an associated p-value that tests whether it significantly affects the outcome. A high p-value suggests that the variable does not contribute much after accounting for other factors.

behind the formula

Regression analysis dates back to the 19th century when Sir Francis Galton, a British polymath, studied hereditary traits in plants. He found that offspring of tall plants tended to be shorter than their parent plants, while offspring of short plants were slightly taller—a phenomenon he called "regression toward the mean." This discovery laid the foundation for modern regression analysis.

"Luck son!"

Example

To answer the question *"Does Gender Really Affect Meal Cost?"*, Lina runs a two-sample t-test comparing male and female meal costs and finds a significant difference ($p < 0.05$). However, when she runs a regression including income level and dining frequency, the p-value for gender becomes insignificant ($p > 0.05$).

Moreover, **confidence intervals** also help in feature selection:

A significant predictor has a confidence interval that does not contain zero

(e.g., [1.2, 3.5]).

feature selection

An insignificant predictor has a confidence interval that includes zero

(e.g., [-1.1, 0.8]),

meaning its effect is uncertain.

This method is widely used in machine learning feature selection to identify meaningful predictors in regression models.

Conclusion:

Regression analysis provides a more complete picture than simple hypothesis tests. It allows researchers to:

✔ Control for multiple factors at once.

✔ Identify confounding variables.

✔ Improve predictive accuracy by selecting the most relevant features.

As Lina's research grows more complex, she realizes that **machine learning** builds upon these statistical foundations, using regression and other models to make data-driven decisions. Understanding regression is the first step toward mastering predictive analytics.

Verify Your Learning!

1. Explain how regression analysis improves upon traditional hypothesis testing when analyzing relationships between variables. In your response, discuss the limitations of univariate hypothesis testing and how regression helps control for confounding variables. Provide an example of a situation where using a regression model would be more appropriate than a simple t-test.
2. A researcher wants to analyze the factors influencing students' monthly transportation expenses. He collects data on students' transportation costs (dependent variable) and considers independent variables such as income level, distance from campus, and mode of transportation. After running a regression analysis, he finds that the p-value for distance is significant ($p < 0.05$), but the p-value for income level is not ($p > 0.05$). How should the researcher interpret these findings? Additionally, explain how confidence intervals for the coefficients can help in deciding which predictors to include in the final model.

Explore the world of statistics with Python through Colab notebook—to get started, access it here: Hypothesis_Testing.ipynb

How This Concept Shapes My Field

A Chilling What-If

Can you imagine what the real world would look like without Hypothesis Testing and Regression Analysis for Hypothesis Testing?

A world without hypothesis testing and regression analysis would be chaotic—decisions would be based on intuition rather than evidence. Medicine, economics, and technology would lack reliable predictions, leading to flawed policies, ineffective treatments, and misguided innovations.

Data gives us the power to see beyond randomness, to find patterns in the noise, and to build a future grounded in reason. Without these tools, we'd be navigating blindly in a storm.

Here's a visual representation of such a world: a blurry, uncertain landscape where patterns remain hidden.

Here is a visual representation of a world without hypothesis testing and regression analysis—chaotic, uncertain, and lacking predictive power. Without these tools, patterns remain hidden, and decisions become guesswork.

Summary

Concept	Key Formula / Notes
Null Hypothesis (H_0)	Assumes no effect or difference.
Alternative Hypothesis (H_a)	Represents what we seek to prove.
Test Statistic	$Test\ Statistic\ = \frac{Observed\ Value - Hypothesized\ Value}{Standard\ Error}$
Decision Rule	○ If p-value < α, reject H_0. ○ If p-value ≥ α, fail to reject H_0.
SE for Population Proportion	$SE\ = \sqrt{\frac{p_0(1-p_0)}{n}}$
Test statistics (Population Proportion)	$test\ statistic\ = \frac{\hat{p}-p_0}{SE}$
SE for Difference in Population Proportions	$SE\ = \sqrt{\hat{p}(1\ -\hat{p})(\frac{1}{n_1}+\frac{1}{n_2})}$
The pooled sample proportion	$\hat{p}\ = \frac{x_1+x_2}{n_1+n_2}$
Test statistics (Difference in Population Proportions)	$test\ statistic\ = \frac{(\hat{p}_1-\hat{p}_2)-(0)}{SE}$
SE for Population Mean	$SE\ = \frac{s}{\sqrt{n}}$
Test statistic for Population Mean	$test\ statistic\ = \frac{\bar{x}-\mu_0}{SE}$

Small sample	(n < 30)
Large sample	(n ≥ 30)
linear regression	$y = w_0 + w_1x_1 + w_2x_2 + \dots + w_nx_n$
A significant predictor	has a confidence interval that does not contain zero
An insignificant predictor	has a confidence interval that includes zero

Glossary

1. **Hypothesis Testing**: A statistical method used to decide whether there is enough evidence to reject a null hypothesis in favor of an alternative hypothesis based on sample data.
2. **Null Hypothesis (H_0)**: A statement that assumes no effect or no difference in a population parameter, serving as the default assumption in hypothesis testing.
3. **Alternative Hypothesis (H_a)**: A statement that contradicts the null hypothesis, representing the effect or difference that the researcher wants to prove.
4. **One-Tailed Test**: A hypothesis test in which the alternative hypothesis specifies that the parameter is either greater than or less than the null hypothesis value.
5. **Two-Tailed Test**: A hypothesis test in which the alternative hypothesis specifies that the parameter is different from the null hypothesis value, without directionality.
6. **Significance Level (α)**: The probability of rejecting the null hypothesis when it is actually true (Type I error), commonly set at 0.05 or 0.01.
7. **Type I Error**: The incorrect rejection of a true null hypothesis, leading to a false positive conclusion.
8. **Type II Error**: The failure to reject a false null hypothesis, leading to a false negative conclusion.
9. **Test Statistic**: A standardized value computed from sample data, used to decide whether to reject the null hypothesis.
10. **p-Value**: The probability of obtaining a test statistic as extreme as the observed value under the assumption that the null hypothesis is true.
11. **Decision Rule**: A rule that specifies the conditions under which the null hypothesis should be rejected or not rejected based on the p-value or test statistic.
12. **Confidence Interval**: A range of values within which a population parameter is expected to lie with a specified probability.
13. **Population Proportion**: The fraction of a population that has a specific characteristic.
14. **Sample Proportion ($\hat{p}$)**: The proportion observed in a sample, used as an estimate of the population proportion.

15. **Standard Error**: The standard deviation of a sample statistic, measuring its variability across different samples.
16. **Pooled Sample Proportion**: A weighted average of sample proportions used in hypothesis testing for comparing two population proportions.
17. **Pooled Variance**: A weighted average of the variances from two or more groups, used when the groups are assumed to have the same variance. It combines data from multiple samples to estimate a common population variance.
18. **Unpooled Variance**: The variance calculated separately for each group, assuming that the variances of the groups are different and not combining them into a single estimate.
19. **Population Mean (μ)**: The average value of a population characteristic.
20. **Sample Mean ($\bar{x}$)**: The average value computed from a sample, used as an estimate of the population mean.
21. **Standard Deviation (σ)**: A measure of the dispersion of values around the mean in a population.
22. **Sample Size (n)**: The number of observations in a sample.
23. **Standard Error of the Mean**: The standard deviation of the sampling distribution of the sample mean.
24. **Linear Regression**: The simplest form of regression, involving a straight-line relationship between the independent and dependent variables.

References

Visual Activity

Hypothesis testing is a fundamental concept in statistics that helps us make informed decisions based on sample data. It allows us to determine whether an observed effect is due to chance or if there is significant evidence to support an alternative claim. Through this simulation, you will learn how to set up null and alternative hypotheses, manipulate sample data, observe test statistics, and interpret p-values. By interacting with visual tools, you will gain a deeper understanding of statistical significance, critical regions, and how different factors—such as sample size and significance level—affect hypothesis test outcomes.

For that purpose, scan this QR code:

Or through this link: [Hypothesis testing – GeoGebra]

How to play

start by selecting a hypothesis testing module and setting up your null and alternative hypotheses. Adjust key parameters such as sample size, population mean, and significance level. Generate sample data and observe the test statistic on the distribution curve. The simulation will provide a p-value, which you can use to decide whether to reject or fail to reject the null hypothesis. Experiment by changing parameters and seeing how different values affect the results, helping you grasp the impact of statistical choices in real-world scenarios.

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